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A Systematic Literature Review of the Practical Applications of Artificial Intelligence-Generated Content (AIGC) Using OpenAI ChatGPT, Copilot, and Codex in Programming Education

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Abstract

This systematic literature review investigated the practical applications of Artificial Intelligence-Generated Content (AIGC) tools, specifically ChatGPT, Copilot, and Codex, in programming education. The review synthesized current research trends and key applications, with a particular focus on large language models (LLMs) and advanced chatbots. The findings revealed a significant increase in research interest following the release of ChatGPT, with most studies concentrating on university-level programming education. The review identified eight primary applications of AIGC tools in programming education, including evaluating AI performance in solving programming tasks, providing AI-driven code generation and assistance, automating assessment review and feedback, delivering personalized learning and tutoring, supporting educators and instructional design, investigating educator and student perceptions, checking for plagiarism, and exploring AI's impact on curriculum design. Despite the promising potential of these tools, critical gaps remained, particularly in primary and secondary education, as well as in online and blended learning environments. Future research was recommended to address these areas and to explore a broader range of AIGC tools to unlock their full potential for enhancing programming education.

CCS Concepts

• **Computing methodologies**; • **Artificial intelligence**; • **Natural language processing**; • **Natural language generation**;

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Keywords

Artificial intelligence generated content, Applications of AIGC, ChatGPT, Systematic literature review, Educational technology, Programming education, Large language models, Chatbots

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1 Introduction

The rise of artificial intelligence (AI) has brought transformative changes across various fields, including education [1]. In particular, the advent of large language models and AI-generated content (AIGC) tools have shown significant potential in programming education [2]. These tools exhibit advanced natural language processing and code generation capabilities, making them valuable resources for enhancing teaching and learning experiences in technical domains.

Programming education poses unique challenges, requiring students to develop conceptual understanding and practical coding skills [3], along with computational thinking [4] [5], positive learning motivation [6], and learning attitudes [7]. Traditional instructional methods often struggle to provide individualized support and timely feedback, which is crucial for mastering programming [8]. AIGC tools, such as ChatGPT, may provide solutions by delivering on-demand explanations, generating code snippets, and providing adaptive learning support tailored to individual student needs [9].

Despite these advantages, integrating AIGC into programming education raises concerns for educators [10] [11]. Issues such as academic integrity, dependency on AI, and the necessity for evolving pedagogical strategies must be addressed to harness these technologies effectively. Given the rapid evolution of AIGC tools, such as

ChatGPT, Copilot, and Codex, educators and researchers must critically evaluate their implications for these innovative technologies.

This systematic literature review aims to synthesize current research on the practical applications and research trends of AIGC tools in programming education. Specifically, we aim to address the following research questions:

1) What are the current research trends regarding AIGC tools in programming education?

2) What are the key practical applications of AIGC tools in programming courses?

By analyzing recent empirical studies and experience reports, this review seeks to provide a comprehensive overview of the current landscape. The insights generated will inform evidence-based approaches for integrating AIGC in programming education and highlight important areas for future research.

After a rigorous selection process, we reviewed 24 relevant papers [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35] from an initial pool of 1173 papers. The remainder of this paper is structured as follows: Section 2 outlines related work. Section 3 details the methodology. Section 4 presents the results, focusing on key applications and trends. Section 5 concludes with a discussion of the findings.

2 Related Work

The rapid advancement of AIGC tools has significantly impacted various fields, leading to several studies on their application in education. Zawacki-Richter et al. [1] conducted a systematic review of 134 articles examining the application of AI in higher education, with a particular emphasis on educational technology and its integration. Their review provided valuable insights into the macro-level trends and theoretical research within AI in education. It identified vital themes such as performance assessment, applications, advantages, and risks associated with AI tools. Similarly, Lo [36] conducted a rapid review of 50 papers, focusing on the initial impact of ChatGPT across various educational disciplines during the three months following its launch. His study provided insights into the effectiveness of ChatGPT in different subject areas, including its satisfactory performance in programming education and its challenges in other fields like mathematics.

These studies highlight essential aspects of AI in education but share common limitations. They focus on broad applications of AIGC across various educational fields without specifically emphasizing programming education. Additionally, Their scope is limited by focusing on early AIGC adoption, missing recent advancements like ChatGPT-4, Copilot, and Codex.

While these studies provide foundational insights into AIGC's role in education, only a few studies have explored its specific application in programming education. We found two review papers related to this specific domain. Deriba et al. [37] provided a mini-review analyzing 13 articles published in 2023, focusing on how ChatGPT can enhance basic programming education. Their review found that ChatGPT can assist in code explanation and handle complex topics. However, their study was presented as a poster presentation, not a full paper, limiting the depth of analysis. On the other hand, Anagnostopoulos [11] also offered a brief overview of ChatGPT's impact on programming education, focusing on applications, advantages, and limitations, such as its use in code management, programming tutoring, and natural language processing. However, his review concentrated on ChatGPT-3.5, without addressing more recent updates to the tool. This limitation reduces the comprehensiveness of his findings about the newer capabilities of ChatGPT-4 and other AIGC tools.

In contrast to these prior reviews, our study covers literature up to 2024, including the latest advancements in AIGC tools. By examining newer developments, we provide a more detailed analysis of AIGC's role in programming education, addressing gaps in the existing research.

3 Methodology

This paper followed Kitchenham's systematic literature review process guidelines [38] for conducting rigorous software engineering and computer science research reviews. The review process consisted of the following key steps.

3.1 Search Strategy

A comprehensive search was conducted across major digital libraries and academic databases, including ACM Digital Library and Web of Science. The search string employed was: ("ChatGPT" OR "GPT" OR "Artificial Intelligence Generated Content" OR "AIGC" OR "generative AI" OR "Large Language Model") AND ("education" OR "school" OR "teach" OR "course" OR "learn" OR "student") AND ("programming" OR "coding" OR "CS1").

3.2 Inclusion and Exclusion Criteria

Given the wide range of retrieved papers, we applied strict inclusion and exclusion criteria to ensure relevance. These criteria focused on studies examining the applications and research trends of integrating AIGC tools in programming education. We developed these criteria using Kitchenham's systematic literature review process guidelines [38]. The inclusion criteria are as follows:

- The paper examines the application of AIGC tools in programming education.
- The study focuses on the applications and research trends of using AIGC tools in programming courses.
- The paper addresses research questions related to the applications and trends of AIGC in programming education.
- Empirical studies, experience reports, or theoretical analyses discuss the impact of AIGC tools on student learning, instructor capabilities, and educational outcomes.
- Studies published in peer-reviewed journals, conferences, or reputable academic sources.

Studies that did not address the integration of AIGC tools in programming education or focused on general computer science education without evaluating the impact of these tools were excluded. The exclusion criteria are summarized below.

- Papers are not focusing on AIGC tools in programming education.
- Studies related to general computer science courses or other subjects without specific emphasis on programming.

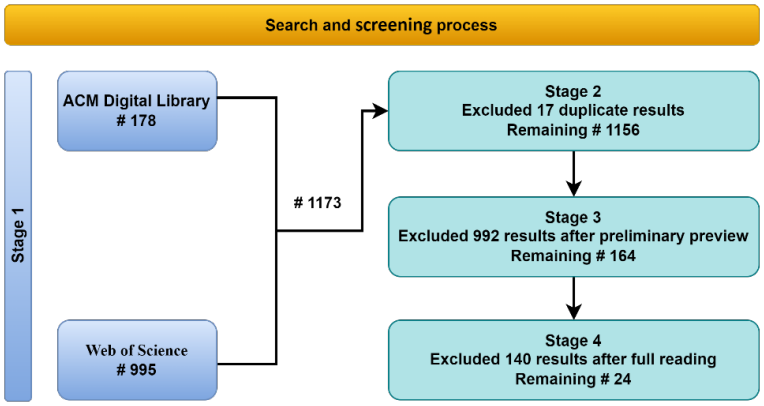

Figure 1: Literature Search and Screening Process

Table 1: Distribution of AIGC Research in Programming Education by Region and Year

Region	2023	Jan-Mar 2024	Percent	Papers
Asia	6	5	34.4%	[12], [18], [20], [21], [22], [23], [26], [27], [28], [30], [31]
America	5	3	25.0%	[14], [15], [16], [18], [19], [25], [31], [33]
Europe	6	2	25.0%	[13], [17], [24], [29], [31], [32], [33], [34]
Oceania	4	0	12.5%	[19], [31], [32], [35]
Africa	1	0	3.1%	[31]

- Papers lacking empirical data, experience reports, or theoretical analysis relevant to the research questions.
- Papers that were not accessible or not written in English.

3.3 Literature Search and Screening Process

We employed a four-stage literature search and screening process to identify the most pertinent studies for subsequent analysis, as depicted in Figure 1.

Stage 1: In March 2024, we utilized a pre-defined search string to search abstracts and full-text papers across two databases. The ACM Digital Library returned 178 results, and the Web of Science returned 995, yielding 1173 results.

Stage 2: We excluded 17 duplicate results based on title, author, and DOI, resulting in 1156 unique results.

Stage 3: To determine paper eligibility, we performed an initial screening. Titles, abstracts, and keywords were manually reviewed to ensure they met the inclusion and exclusion criteria. A text search was conducted to find paragraphs with key phrases, which were then analyzed to exclude non-relevant papers. This process excluded 992 papers, leaving 164 papers.

Stage 4: The remaining papers were fully read, and the inclusion and exclusion criteria were re-evaluated. Relevant information was extracted from the papers, including title, objectives, intervention description, main findings, educational context, and quality data. This led to the exclusion of 140 studies, resulting in 24 relevant papers thoroughly analyzed for the literature review.

4 Results

4.1 Publication Region and Year

The research on AIGC applications in programming education has significantly increased since ChatGPT's public release in November 2022. Our systematic literature review reveals a concentrated focus on this topic in 2023 and early 2024, indicating an accelerating trend in research output, as shown in Table 1.

Geographically, the research distribution shows a global interest, with Asia leading at 34.4% of the publications, followed by Europe and America, each contributing 25%. Oceania accounts for 12.5%, while Africa contributes 3.1%, suggesting an emerging interest. It's important to note that while there are 24 unique papers in total, some studies involve students from multiple regions, resulting in individual papers being counted in more than one geographical category.

This rapid increase in publications indicates exponential growth in the field, driven by the increasing accessibility of AIGC tools and their transformative potential in programming education. Asia's prominence may stem from its strong emphasis on education, supported by significant technological investments and government research initiatives. These factors may facilitate rapid adoption and innovation, contributing to Asia's substantial research output. As research evolves, we expect more diverse contributions and deeper exploration of AIGC applications in programming education.

Table 2: Distribution of AIGC Research in Programming Education by Educational Level and Type

Educational Type	Traditional	Online	Blended	Percent	Papers
University	21	2	0	95.8%	[12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [31], [32], [33], [34], [35]
Others	0	0	1	4.2%	[30]

Table 3: Distribution of AIGC Tools in Programming Education Research

AIGC Tools	Number of Studies	Percent	Papers
ChatGPT-3/3.5	17	58.6%	[14], [15], [16], [17], [19], [20], [21], [22], [24], [25], [26], [27], [29], [30], [33], [34], [35]
ChatGPT-4	5	17.2%	[15], [18], [26], [27], [33]
ChatGPT (No version)	5	17.2%	[12], [13], [23], [28], [31]
Copilot	1	3.4%	[31]
Codex	1	3.4%	[32]

4.2 Educational Level and Type

Analyzing AIGC applications in programming education reveals significant disparities across educational levels and types, as detailed in Table 2. At the educational level, an overwhelming 95.8% of studies focus on university settings, with only one study (4.2%) not specifying the level, categorized as others. This distribution highlights a substantial research gap in primary and secondary education.

Regarding the educational type, traditional classroom environments dominate the research landscape, accounting for 21 out of 24 studies. Only two studies explore online teaching environments, and one investigates a blended approach. This distribution underscores a critical research gap in online and blended learning contexts for AIGC applications in programming education.

These findings suggest a need for diversification in AIGC research across educational levels and types. Future studies should address the underrepresented areas, particularly in primary and secondary education and online and blended learning environments, to provide a more comprehensive understanding of AIGC's potential in programming education across various contexts.

4.3 Distribution of AIGC Tools

As detailed in Table 3, the analysis of AIGC tools used in programming education reveals a clear dominance of ChatGPT, particularly versions 3 and 3.5, which account for 58.6% of the studies. This prevalence may be attributed to their widespread availability and established capabilities. ChatGPT-4, despite its enhanced performance, was used in only 17.2% of the studies, likely due to its recent release and limited access. Notably, 17.2% of the studies did not specify the ChatGPT version used, highlighting a need for more precise reporting in future research.

The relatively low representation of Copilot (3.4%) and OpenAI Codex (3.4%) in the studies is noteworthy, considering their specific design for coding tasks. This underutilization suggests potential opportunities for future research to explore these tools' efficacy in programming education contexts.

Interestingly, some studies employed multiple AIGC tools concurrently. For instance, Phung et al. [33] showcased a novel dual-model tutoring technique, utilizing ChatGPT-4 for generating programming hints and ChatGPT-3.5 for validation. This approach demonstrates the potential for using different AI models to enhance programming education.

It is important to note that the rapid evolution of AIGC tools may outpace academic research. Recent models, such as Claude, Gemini, and Meta LLama, were not represented in the reviewed literature, indicating a gap between technological advancements and their integration into educational research. Future studies should consider exploring these newer models.

4.4 Practical Applications of AIGC in Programming Education

The examination of practical applications of AIGC in programming education reveals varying levels of research focus across different areas, as shown in Table 4.

The most explored application, accounting for 23.3% of the studies, evaluates AI's performance in solving programming tasks. AI-driven code generation and assistance follow closely with 20.9%, and assessment review and feedback automation at 16.3%. Less explored but equally essential areas include personalized learning and tutoring with AI at 11.6%, support for educators and instructional design at 9.3%, and investigations into educator and student perceptions at 9.3%. Plagiarism checking and exploring AI's impact on curriculum design, each at 4.7%, represent research gaps that warrant further investigation to harness AI's potential in educational settings fully. The following sections detail these findings, incorporating data and specific examples from the reviewed papers.

4.4.1 Evaluating AI Performance in Solving Programming Tasks. This area garnered significant attention, comprising 23.3% of the reviewed studies. The research focused on how AI models like ChatGPT, ChatGPT-4, and Codex performed relative to human students. For instance, Piccolo et al. [19] evaluated ChatGPT's effectiveness

Table 4: Distribution of Research Focus Areas in AIGC Applications for Programming Education

Application Type	Percent	Papers
Evaluating AI Performance in Solving Programming Tasks	23.3%	[13], [14], [15], [17], [19], [25], [26], [29], [34], [35]
AI-Driven Code Generation and Assistance	20.9%	[13], [15], [16], [18], [22], [26], [27], [29], [30]
Assessment Review and Feedback Automation	16.3%	[13], [14], [15], [21], [24], [28], [34]
Personalized Learning and Tutoring with AI	11.6%	[18], [21], [24], [30], [33]
Supporting Educators and Instructional Design	9.3%	[17], [27], [31], [32]
Investigating Educator and Student Perceptions	9.3%	[12], [20], [23], [31]
Plagiarism Checking	4.7%	[14], [28]
Exploring AI's Impact on Curriculum Design	4.7%	[31], [35]

in solving programming exercises within an introductory bioinformatics course, achieving a 75.5% success rate on the first attempt and 97.3% after iterative feedback. Savelka et al. [25] highlighted ChatGPT-4's partial success across three Python courses, scoring above 55% in more accessible modules but failing more challenging assessments. Additionally, Finnie-Ansley et al. [35] tested Codex on advanced programming tasks from a CS2 course, where it often outperformed human students. These findings suggested significant pedagogical implications for integrating AI into undergraduate computing education. This high level of interest underscored the importance of AI as a valuable tool in assessing and enhancing students' programming skills. Yet, it also highlighted the need for more nuanced research to optimize AI's capabilities in this domain.

4.4.2 AI-Driven Code Generation and Assistance. With 20.9% of the studies focusing on this area, AI-driven code generation and real-time assistance demonstrated a transformative shift in programming education. Da Silva et al. [16] explored ChatGPT's role in generating code snippets and offering programming hints, with a Likert scale questionnaire among 40 Brazilian students indicating strong support for ChatGPT's educational potential. Kuramitsu et al. [22] detailed KOGI, a system incorporating ChatGPT within Jupyter environments, significantly reducing unresolved student errors in Algorithms and Data Science courses. Wu et al. [30] described an AI-based platform providing personalized learning experiences, addressing the shortage of qualified teachers, and customizing the educational process. The consistent positive outcomes from these studies indicated that AI-driven assistance could significantly improve learning efficiency and effectiveness. However, more research was necessary to address potential challenges and optimize these systems further.

4.4.3 Assessment Review and Feedback Automation. Studies in this area, representing 16.3% of the total, have shown encouraging outcomes. Balse et al. [21] explored ChatGPT-3's capacity to provide feedback on 1211 student code submissions, with correctness checks achieving 57% to 79% accuracy, critiques from 41% to 77%, and suggestions scoring between 32% and 93%. Azaiz et al. [24] used ChatGPT-3.5 to provide automated feedback on programming assignments, identifying 73% of submissions as correct or incorrect and generating effective feedback in 59% of cases. These findings highlighted the potential of AI to enhance the efficiency and quality of assessment processes. However, the variability in accuracy underscored the need for continued refinement and testing to ensure that AI feedback could reliably meet educational standards.

4.4.4 Personalized Learning and Tutoring with AI.. Comprising 11.6% of the studies, personalized learning and tutoring with AI profoundly impacted teaching methodologies. Jin and Kim [18] introduced a GPT-empowered eLearning system for programming languages that dynamically generated personalized content and managed eLearning sessions. Phung et al. [33] showcased a dual-model tutoring technique where ChatGPT-4 generated programming hints and ChatGPT-3.5 validated them, integrating symbolic information from failing test cases to enhance instructional quality. These advancements illustrated AI's potential to offer tailored educational experiences, though further exploration was required to address the individual learning needs comprehensively and effectively integrate AI into diverse educational contexts.

4.4.5 Supporting Educators and Instructional Design. Supporting educators and instructional design was addressed in 9.3% of the studies. Bringula [27] documented ChatGPT's use in facilitating teaching-learning activities, creating class materials, and designing assessments in a Data Analytics course. While effective, limitations included generating incorrect reference materials and producing pattern-like content. Sarsa et al. [32] highlighted OpenAI Codex's ability to create programming exercises and explanations, making novel and ready-to-use content that influenced programming concepts and thematic contexts. These examples showed that while AI could significantly aid educators, there was a crucial need for ongoing improvements and validations to ensure AI-generated content's accuracy and pedagogical value.

4.4.6 Investigating Educator and Student Perceptions. Understanding perceptions of AI in programming education was critical, covered in 9.3% of the studies. Jing et al. [12] used a quasi-experimental design to examine factors like AI literacy and programming knowledge base that affected learners' use of AI tools. Husain [23] explored programming instructors' perspectives, revealing a spectrum of enthusiasm and concern regarding AI's educational role. These studies emphasized the importance of gauging and addressing the sentiments of both educators and students toward AI tools to ensure smoother integration.

4.4.7 Plagiarism Checking. In 4.7% of the studies, plagiarism checking was crucial for maintaining academic integrity. Karnalim et

al. [28] developed a method using 34 features to detect AI-assisted code submissions, achieving an 89% precision rate. Ellis et al. [14] focused on instructors' ability to identify AI-generated Python programming homework, highlighting the challenges in distinguishing AI-generated work from student submissions. This research gap indicated the need for more robust and sophisticated detection methods to keep up with advancements in AI-generated content and ensure fair academic practices.

4.4.8 Exploring AI's Impact on Curriculum Design. Also covered in 4.7% of the studies, the impact of AI on curriculum design was profound. Lau and Guo [31] reported that university programming instructors were rethinking courses in response to AI code-generation tools. Interviews with instructors from nine countries revealed varied integration strategies. Finnie-Ansley et al. [35] documented Codex's superior performance on advanced CS2 programming exercises compared to human students, suggesting a pressing need to adapt curricula for an AI-augmented future. This underexplored area highlighted a significant opportunity for future research to develop adaptive curricula that better prepare students for an AI-driven educational landscape.

5 Conclusion and Discussion

This review synthesized current research on AIGC tools in programming education, revealing their rapid integration and potential impact. A sharp rise in ChatGPT-related studies was noted, especially in 2023 and early 2024, reflecting a growing recognition of its transformative role. The geographical distribution of these studies highlighted a global interest, with Asia leading the research output, followed by Europe and America. This widespread interest underscored the universal applicability and relevance of AIGC tools in diverse educational contexts.

Most studies focused on the university level, revealing a research gap in primary and secondary education. Furthermore, traditional classroom environments dominated the research landscape, with limited attention to online and blended learning environments. These findings suggested the need for a more diversified research approach that includes various educational levels and teaching modalities to fully understand the potential of AIGC tools across different learning environments.

In addition, ChatGPT, particularly its versions 3 and 3.5, emerged as the most frequently studied AIGC tool, likely due to its widespread availability and established capabilities. Although the newer ChatGPT-4 showed promise with enhanced performance, it needed to be more regularly analyzed, possibly due to its recent release and limited access. The review also noted the underutilization of other specialized AIGC tools, such as Copilot and OpenAI Codex, highlighting an area for future research to explore the efficacy of these tools in programming education.

The practical applications of AIGC tools in programming education were diverse, with significant attention given to evaluating AI performance in solving programming tasks, AI-driven code generation and assistance, and assessment review and feedback automation. While these tools show potential to improve learning, the variability in their performance indicates a need for further refinement.

Despite the promising applications, integrating AIGC tools raised essential concerns that must be addressed. Issues such as academic integrity and the necessity for evolving pedagogical strategies were critical considerations for effectively harnessing these technologies. Ensuring that AI tools are used ethically and effectively in educational settings is paramount to their successful integration.

Additionally, while evaluating AI performance in solving programming tasks received the most attention, areas such as personalized learning and tutoring with AI, supporting educators and instructional design, and investigating educator and student perceptions were less explored. These areas are crucial for a holistic understanding of AIGC tools' impact and should be prioritized in future research.

In conclusion, while AIGC tools showed potential in transforming programming education, there is a need for more comprehensive research across various educational levels. Future studies should address the identified gaps, particularly in primary and secondary education and online and blended learning environments. Global studies and further exploration of specialized tools will enhance AI's role in programming education. By addressing these areas, educators can develop evidence-based approaches that maximize the benefits of AIGC tools, ensuring they contribute positively to education.

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